TOMATO – Technical Optimization of Multi-objective Agricultural Target Outcomes

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Abstract

Controlled-environment agriculture offers precise control over crop growth, but its practical viability depends on optimizing yield under tight resource and operating constraints. We formulate indoor crop planning as a constrained surrogate-based optimization problem over greenhouse control and resource variables to maximize crop yield. To do so, we implemented a unified framework to compare multiple black-box optimizers across four different crops. Yield is evaluated using a CatBoost surrogate, and each crops is modeled as a 14-dimensional continuous design problem with expert-derived bounds and constraints. Within this common interface, we compared constrained Bayesian optimization, genetic algorithm, particle swarm optimization, and CMA-ES with the same feasibility handling.

1. Introduction

As the world population increases and climate becomes more unpredictable, ensuring the security of our food supply is ever more critical. Luckily, new methods of vertical, indoor farming are emerging as potential solutions which allow us to better control plant-growing environments and increase output in dense urban areas. However, indoor growing environments require huge amounts of energy and struggle with business feasibility, particularly as the grid is burdened by other high-energy requirements such as data centers. Therefore, optimizing crop inputs for yield and harvest quality of indoor-farmed crops is crucial and urgent to the success of this strategy. In this paper, we examine this problem by applying and analyzing several optimization algorithms. We fit a black-box, CatBoost model for our plant growth. Under several constraints such as known growing conditions, cost, and land availability, we then optimize over several factors such as days to maturity, photoperiod hours, light intensity, fertilizer usage, and water. We maximize crop yield for four different crops: tomato, cucumber, lettuce, and pepper.

2. Literature Review

Indoor farming began to grow in popularity throughout the 2000s, advocated as a sustainable farming approach. In 2022, indoor farming attracted $2.4 billion in funding from governments and investors seeing massive potential in this emerging technology (Haitao Li, 2025). However, as of 2025, controlled environment agriculture provides less than 1% of US food crops while consuming more energy than all open-field cultivation (Mills, 2025). Many vertical farming startups such as AeroFarms have failed to scale or turn profits, falling prey to fundamental inefficiencies in crop optimization and high operating costs. In a study conducted by the World Economic Forum, there are several key recommendations for optimizing the vertical farming industry and improving the economics of these startups (Haitao Li, 2025). We will focus on two: implementing resource planning and increasing crop yields.

Research into this optimization problem is still developing. An article published by Daniels et. al in 2023 demonstrated the importance of optimal control within indoor farming. The authors examined wheat growth using an optimal control approach with two main objectives (Annalena Daniels, 2023). First, they optimized inputs such as water and temperature. They also optimized duration of plant's growth period. Daniels and co-authors employed a nonlinear, discrete-time hybrid model to demonstrate the feasibility of control theory in vertical farming. They present two specific optimization models for this problem, SIMPLE-S and SIMPLE-SC, demonstrating that their models result in a 25% energy cost reduction and increase in harvest per year (Annalena Daniels, 2023). In our paper, we expand upon their inputs and constraints to further optimize over factors such as fertilization and soil quality. We also optimize explicitly for yield, building a CatBoost-based surrogate model which allows for a continuous-time model to better model real-world scenarios. The findings of this paper are reflected in yearly advances, largely through the Autonomous Greenhouse Challenge (University and Research, 2025). This challenge brings together teams of researchers around the globe annually, encouraging teams to find optimal inputs for tomatoes, lettuce, and cucumbers using novel optimization algorithms for crop management. These teams have found that managing heat, lighting, carbon dioxide levels, irrigation, and fertilization is critical to crop success, lending support to our set of inputs in our optimization problem formulation.

Another influential paper published in 2024 by Kaiser et. al studied optimization of resource use for vertical farming. They examined important dials to plant growth including photoperiod, air flow, carbon dioxide exposure, and temperature. They write that dynamic changes in these inputs can keep operational costs low while still leading to good quality and yield outcomes for plants (Elias Kaiser, 2024). They suggest a continuous feedback loop for crop management, constantly adjusting inputs to respond to market prices while keeping plants healthy. This continuous feedback loop has been implemented in real farms, such as a new farm in Virgina (Nguyen, 2024). However, their research does not present the best optimization algorithm to use for these plants nor discuss the target level for optimal yield beyond staying within a certain growing range (for example, a temperature band). Our paper builds upon their analysis by presenting a concrete set of inputs to target for optimal plant growth, though we acknowledge that staying within a certain input's "reasonable" band may improve economic feasibility for new startups. We also provide recommendations for optimization algorithms that can be used for these continuous crop management systems.

While existing optimization research into vertical farming is limited, the literature above serves as a promising base for our project and highlights the importance of this work. Understanding how we can optimize plant growth is critical to the success of the industry, and there is still much to be done regarding determining the best inputs under economically-feasible constraints.

3. Approach and Methodology

Unfortunately, it is not possible to grow thousands of plants over the course of this class to test our optimization algorithms. However, we have the power of statistics, computer science, and machine learning on our hands. Thus, to model the growth behavior, we use a surrogate model trained on existing, open-source datasets which include real-world data on crop inputs and growth outcomes (Khan, 2026). Our surrogate model becomes our simulated-growth system. Before training a surrogate model, we clean our data for missing entries or duplicates. We then conduct feature engineering, adding features such as VPR. We test several surrogate models and choose the model with the best test RMSE, covered in more depth below.

We now have a constrained optimization problem. Our constraints are covered in depth in our Appendix A and include 14 crop-specific bounds. Examples of these constraints include tomato photoperiod hours being restricted to between 12 and 16, based on existing research on tomato chlorosis. Days to maturity is bounded by the 5th and 95th percentile of the training data per crop as realistic ranges from real growing data, as we want to ensure our crops are fully grown yet can be harvested as soon as possible. We have 8 coupling constraints, which make the feasible region non-rectangular. These coupling constraints include Vapor Pressure Deficit bounds between 0.5 kPa and 1.2 kPa, due to known risks of fungal disease and plant water stress. Vapor Pressure Deficit is calculated using humidity and temperature:

$$\mathrm{VPD}(T, H) = \left(1 - \frac{H}{100}\right) \cdot 0.6108 \exp\left(\frac{17.27T}{T + 237.3}\right) \in [0.5, 1.2] \text{ kPa}.$$

Additionally, we include an electricity budget of 300 kWh/m^2/cycle and a water budget of 400 L/m^2/cycle. We also include an operating-cost cap of \$300 m^2/cycle. Our land budget is 3 acres per plot, with 10 plots stacked vertically based on existing farms (Nguyen, 2024).

For our optimization problem, we choose four different optimization methods. These methods were selected for their ability to optimize black-box problems, as we do not have a nice or defined objective function to optimize over. These four optimization algorithms are described below:

- **Bayesian Optimization**: This stochastic method treats the surrogate as a black-box function and explores promising regions using a probabilistic model (e.g. Gaussian process). To apply this, we would iteratively choose a small set of sampled greenhouse configurations, fit the probabilistic surrogate model to predict expected yield, then choose the next configurations that maximize expected improvement. This is most effective when evaluations are expensive.

- **Genetic Algorithm**: This is a population method with greenhouse conditions as individuals. We would iteratively evaluate each setup using the surrogate model and select the best individuals, then apply crossover and mutation. This typically works well for multiobjective optimization.

- **CMA-ES**: This genetic algorithm involves fitting a multivariate Gaussian distribution to our greenhouse parameters, sampling candidate solutions, and evaluating with the surrogate model. We then update our Gaussian based on the best candidates and repeat. This works especially well for continuous black-box optimizations.

- **Particle Swarm Optimization**: This algorithm represents the greenhouse parameters as particles. Initialized with random positions and velocities in the feasible parameter space, we keep track of each particles' personal best position as well as the global best position in the swarm, and update velocities and positions to move closer to the global best. Constraints are enforced by clipping infeasible particles or applying penalties. This method often converges faster and works well for continuous optimization problems.

Finally, to analyze the results of our algorithms, we compare the crop yield of each algorithm. We also compare the time requirements, constraint violation rate, and sample efficiency for each algorithm. Additionally, we also test robustness and stability by running the stochastic algorithms multiple times with different seeds. In our Results and Analysis section, we also evaluate why certain algorithms worked better for this problem space by examining the specific mechanisms of the high-performing algorithms. Since we have multiple objectives, we also build a Pareto-Optimal curve that shows the trade-offs between energy usage, cost, and growth. We also compare the number of Pareto-optimal solutions found in potential solution to our multiobjective problem. In our analysis below, we include a sensitivity analysis over our algorithms, which crop inputs matter most to each of our models.

4. Results and Analysis

Our surrogate model results are available in the following table. Seeing that the CatBoost model with engineered features had the highest Test R^2 of 0.8308, we chose CatBoost for our plant surrogate.

Model Name	**Test RMSE**	**Test R^2**
HGB + engineered features	2.0648	0.8282
CatBoost + engineered features	2.0487	0.8308
XGBoost + engineered features (sanity)	2.0783	0.8259

Table 1: Model performance comparison on test set.

Then, after implementing the above four algorithms, we calculated the median yield over ten seeds for each algorithm for each crop. We also calculated the median wall clock time to run a particular optimization algorithm for a seed, once the algorithm has been tuned for optimal hyperparameters. The results are available in the below tables.

The best hyperparameters for the genetic algorithm provide insight into the type of optimization problem we are optimizing over. The tuned hyperparameters include a population size of 200, pc of 0.8, pm of 2 per dimension, η_c of 2.0, and η_m of 5.0. This indicates that larger populations dominate at our 5000-eval budget, and more mutation helps since we explore more of our large-dimensional problem. A lower SBX distribution index is also helpful, as children will move farther than parents and increase exploration. Altogether, each winning hyperparameter indicated more exploration is better since the genetic algorithm is limited by how much of the 14-dimensional feasible region that it explores. We can explore aggressively on our cheap CatBoost surrogate, however, with a more expensive objective, we would likely see the optima η values shift higher for more exploitation.

Table 2: Median Yield (kg/m^2) per Algorithm

Algorithm	Tomato	Cucumber	Lettuce	Pepper
Bayesian Optimization	23.22	16.84	8.47	12.05
Genetic Algorithm	23.70	17.89	9.32	13.54
CMA-ES	**24.49**	**18.06**	**9.48**	**13.81**
Particle Swarm	24.03	18.01	9.18	13.62

Table 3: Median Time per Algorithm (s)

Algorithm	Tomato	Cucumber	Lettuce	Pepper
Bayesian Optimization	175.1	157.6	172.6	142.1
Genetic Algorithm	**1.39**	**1.73**	**1.82**	**1.68**
CMA-ES	22.0	21.7	20.3	23.5
Particle Swarm	10.2	9.43	10.5	10.1

Under the current cheap-surrogate regime and stopping rules, the population-based methods achieve the strongest final yields. CMA-ES attains the highest mean yield on tomato and lettuce while PSO performs best on cucumber and pepper. This is for several reasons. First, the objective function is relatively cheap, as previously mentioned, and bumpy. Population methods get thousands of evaluations for free, whereas BO struggles to optimize over surface that is piecewise-constant with axis-aligned steps and flat plateaus. Additionally, Figure 8 in the Appendix demonstrates that cost is dominated by electricity. This means that the best feasible points are pinned against the electricity-budget face, and the optimum is not on the interior of the box. Since CMA-ES adapts the full covariance matrix, CMA-ES can rotate and elongate its search distribution to lie along the constraint ridge and stretch towards the optimal corner of the space. Population methods allow the surviving population to hug the boundary, and draws aren't wasted in infeasible territory. PSO also performs well here because particles can slide along the feasible boundary rather than bounce off of it. Finally, the shape of the problem, which is 14-dimensional with variables spanning several orders of magnitude, lends strength to the CMA-ES and PSO methods. In particular, CMA-ES scales to a normalized unit cube. Further analyses of the algorithm performance are availabe in the Appendix.

GA is the fastest method overall, followed by PSO and CMA-ES, while bayesian optimization is slowest due to repeated Gaussian process refitting. GP fitting is $O(n^3)$ and maximizing the constrained-EI with multi-start L-BFGS-B is also slow, which accounts for the time differential between BO and the other methods. This performance difference is due to GA using a batched surrogate call for an entire generation, while CMA-ES and PSO do not use batched population evaluations. CMA-ES and PSO also both run bisection repair, whereas GA avoids the per-candidate constraint-check loop. However, BO is also the most sample-efficient method in terms of surrogate evaluations, not requiring the thousands of evaluations used by population methods. We note that due to the time requirements of Bayesian Optimization, BO was capped at about 200 evaluations while the population methods use thousands of evaluations.

Finally, we present the optimal decision vectors for each algorithm. For comparison, we choose seed 0 for all algorithms and specifically present the results for tomatoes. Across both bayesian optimization and the genetic algorithm, we see that light is at or near the upper bound for all crops as the strongest yield driver. For tomatoes, a warm season crop, we also have photoperiod at the lower bound, indicating that the electricity-budget pressure forces short days with high lighting. Our algorithms also found that high levels of nitrogen increase yield. However, bayesian optimization and the genetic algorithm diverge on fertilizer K, with bayesian optimization choosing the max fertilizer amount while the genetic algorithm chooses a more moderate number. This indicates that the surrogate's K surfaces are flat enough near the optimum that both ridge values give similar yields. As algorithms, bayesian optimization snaps to bounds more aggressively, while the genetic algorithm optimization finds interior values that work well.

Table 4: Variable Values per Algorithm

Algorithm	days to maturity	avg temp C	min temp C	max temp C	humidity %	co2 ppm	light lux	photoperiod h	irrigation mm	fert N	fert P	fert K	pest severity	soil pH
Bayesian Optimization	61.00	25.04	22.05	32.00	70.34	1187.24	39913.05	12.00	5.11	200.00	72.46	300.00	0.00	5.80
Genetic Algorithm	63.38	24.75	21.64	30.23	74.44	1152.99	37967.09	12.15	5.22	197.61	85.90	197.21	0.12	6.32
CMA-ES	61.00	24.64	21.31	28.37	71.80	1009.12	39968.65	12.20	4.97	182.16	79.07	266.14	0.00	6.24
Particle Swarm	61.31	23.88	22.18	26.33	71.48	802.09	35698.49	13.64	4.99	200.00	74.19	252.12	0.00	6.26

Another analysis we ran is a sensitivity analysis over each decision variable near the optimum. We perturb each variable individually by -10%, -5%, 0%, 5%, and 10% of it's bound range. We conduct a standard one-at-a-time perturbation with the other 13 factors held fixed, scoring sensitivity as the range of yield across the 5 perturbations. We see in 5 that light intensity is the most sensitive input, followed by the amount of water that the crop receives. Soil pH is the least sensitive among the 14 crop inputs.

Table 5: Local sensitivity score by crop

Variable	Tomato	Cucumber	Lettuce	Pepper
days_to_maturity	0.270	0.204	0.131	0.065
avg_temperature_C	0.303	0.146	0.200	0.254
min_temperature_C	0.476	0.395	0.311	0.513
max_temperature_C	0.385	0.352	0.512	0.498
humidity_percent	0.084	0.065	0.043	0.115
co2_ppm	0.102	0.123	0.033	0.109
light_intensity_lux	1.382	0.459	0.086	0.716
photoperiod_hours	0.180	0.251	0.029	0.081
irrigation_mm	0.569	0.623	0.184	0.225
fertilizer_N_kg_ha	0.300	0.432	0.132	0.340
fertilizer_P_kg_ha	0.039	0.016	0.120	0.059
fertilizer_K_kg_ha	0.162	0.167	0.105	0.639
pest_severity	0.170	0.221	0.143	0.065
soil_pH	0.010	0.052	0.006	0.013

We also see from the empirical Pareto frontier that the population-based methods often contribute more of the high-yield tradeoff points than Bayesian Optimization, although no single algorithm dominates the entire frontier. Generally, the BO frontier appears the weakest and indicates worse performance across the optimization space.

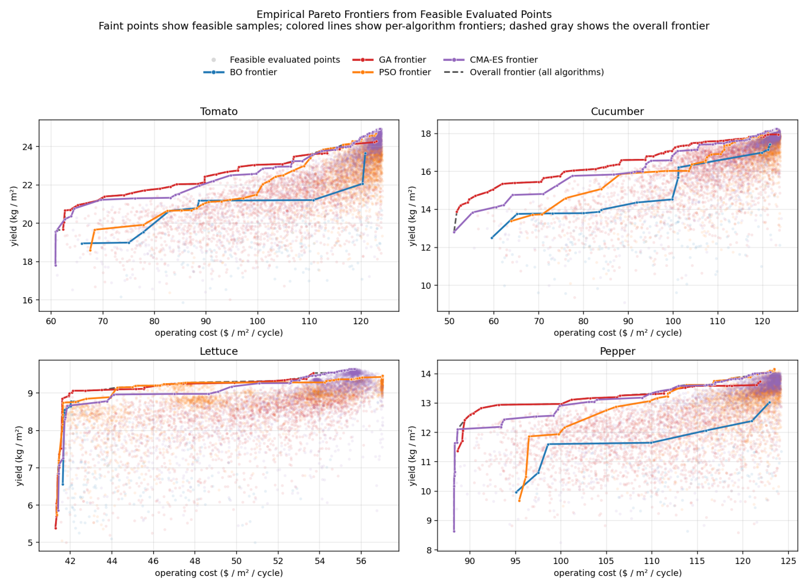

Figure 1: Empirical Pareto Frontier

5. Conclusions

In this work, we formulated indoor crop planning as a constrained surrogate-based optimization problem over 14 greenhouse control and resource variables, and compared Bayesian Optimization, a Genetic Algorithm, Particle Swarm Optimization, and CMA-ES across tomato, cucumber, lettuce, and pepper. Using a CatBoost surrogate for fast yield evaluation, we were able to search a constrained design space defined by agronomic, environmental, and operating-cost limits. Across ten seeds, the population-based methods produced the strongest median yields overall, with CMA-ES performing best on tomato and lettuce and Particle Swarm performing best on cucumber and pepper, while the Genetic Algorithm was by far the fastest in wall-clock time across all crops.

These results suggest that in a cheap-objective regime, where surrogate evaluations are fast and large numbers of candidate solutions can be explored, population-based search methods are especially effective for high-dimensional constrained greenhouse optimization. Our sensitivity analysis further showed that light intensity and irrigation are among the most influential variables near the optimum, while the empirical Pareto frontier highlighted the tradeoff between yield and operating cost. At the same time, Bayesian Optimization remained more sample-efficient in terms of the number of evaluations used, so its advantages may become more important when objective evaluations are expensive. Future work should compare all methods under matched evaluation budgets, extend the framework to true multiobjective optimization, and validate the resulting operating points against physical greenhouse data.

6. Contributions

Each team member contributed equally to the project work. Both team members worked equally on the project proposal, milestone, and final paper. They each implemented two algorithms for the optimization portion and collaborated for the remainder of the technical work, including the surrogate model and the problem formulation.

Appendix A. Crop-Specific Greenhouse Constraints

Constraint ranges were adapted from published greenhouse guidelines for optimal commercial crop growth conditions.

Table 6: Crop-Specific Greenhouse Constraints

Constraint	**Tomato**	**Cucumber**	**Lettuce**	**Pepper**
Minimum Temperature (°C)	15	18	10	16
Maximum Temperature (°C)	32	35	27	35
Relative Humidity (%)	60–75	70–90	50–70	50–70
CO_2 Concentration (ppm)	800–1200	800–1200	600–1000	800–1200
Light Intensity (lux)	20,000–50,000	20,000–40,000	10,000–20,000	25,000–50,000
Photoperiod (hours/day)	12–16	12–16	10–14	12–16
Irrigation (mm/day)	4–8	5–10	2–5	4–8
Soil pH	5.8–6.8	5.5–7.0	6.0–7.0	5.8–6.8
Nitrogen (kg/ha)	100–200	80–180	50–120	100–180
Phosphorus (kg/ha)	40–100	40–90	30–80	40–100
Potassium (kg/ha)	150–300	120–250	80–180	150–300
Maximum Pest Severity Index	3	3	2	3

Appendix B. Additional Experimental Results

B.1 Cross-Algorithm Performance

Figure 2 summarizes the final best-feasible yield achieved by each method across the four crops.

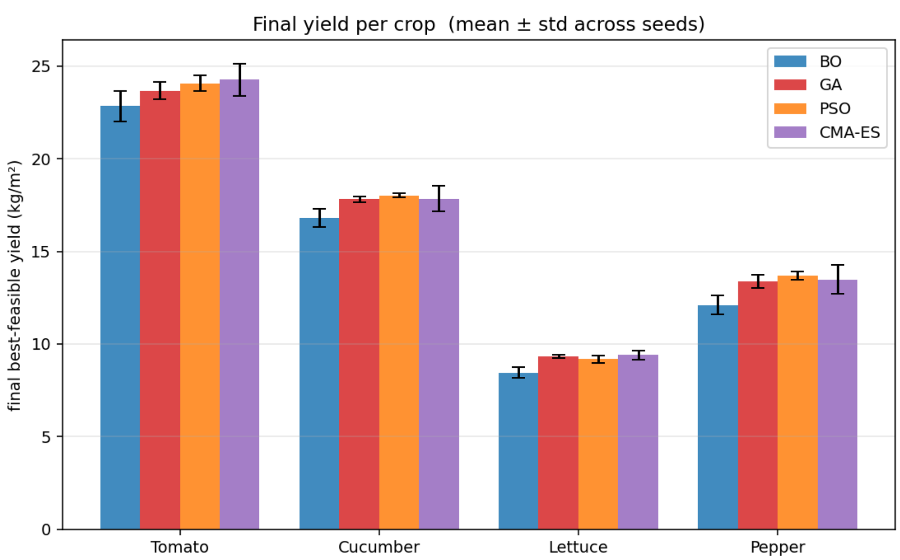

Figure 2: Final best-feasible yield per crop for all four algorithms. Bars show the mean across seeds and error bars show the standard deviation. CMA-ES achieves the highest final yield on tomato and lettuce, while Particle Swarm performs best on cucumber and pepper.

Figure 3 complements this by showing the number of surrogate evaluations required to reach 95% of the best-found yield.

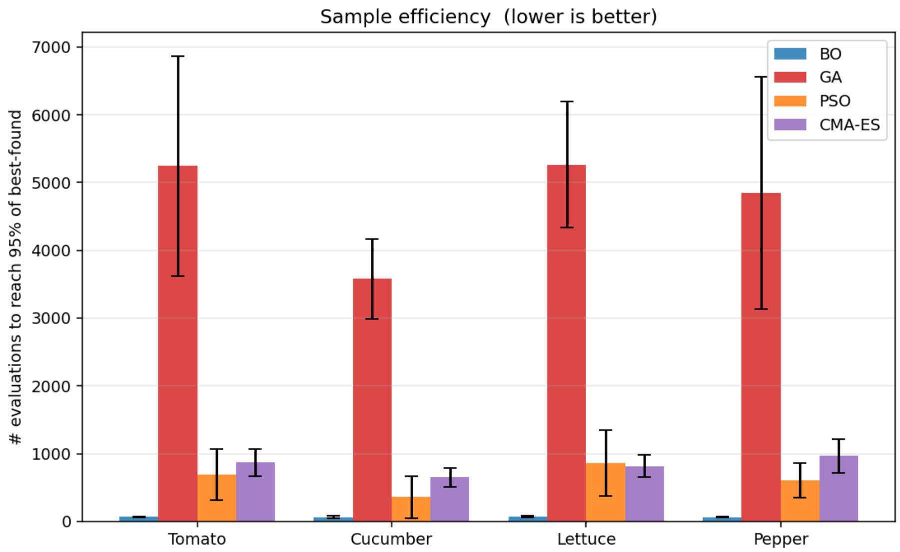

Figure 3: Sample efficiency measured as the number of surrogate evaluations required to reach 95% of the best-found yield. Lower is better. Bayesian Optimization is the most sample-efficient method, while the population-based methods require substantially more evaluations to reach their strongest solutions.

A representative convergence trace for tomato is shown in Figure 4.

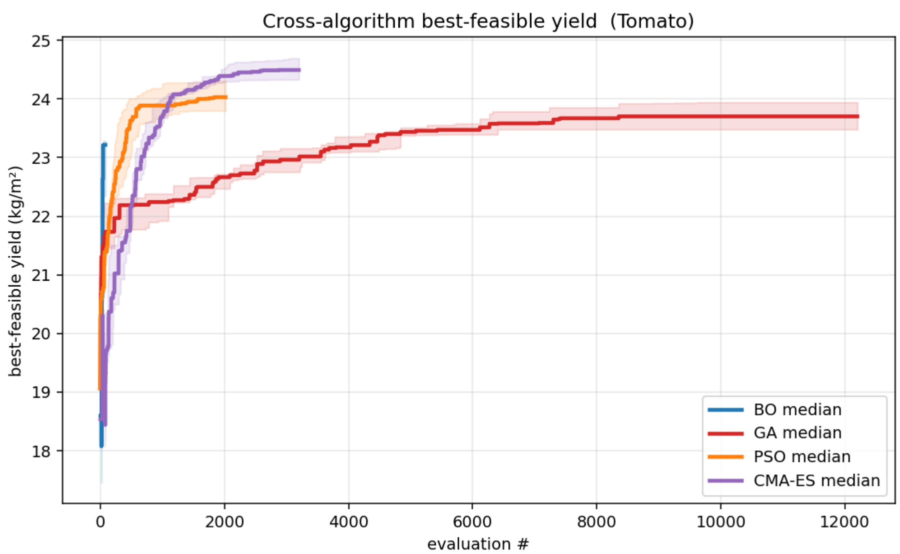

Figure 4: Representative convergence behavior on tomato. Bayesian Optimization improves rapidly at very low evaluation counts but plateaus early, while the population-based methods continue improving and eventually attain higher best-feasible yields.

B.2 Representative Hyperparameter Sweep

Figure 5 shows one representative hyperparameter sweep for the Genetic Algorithm.

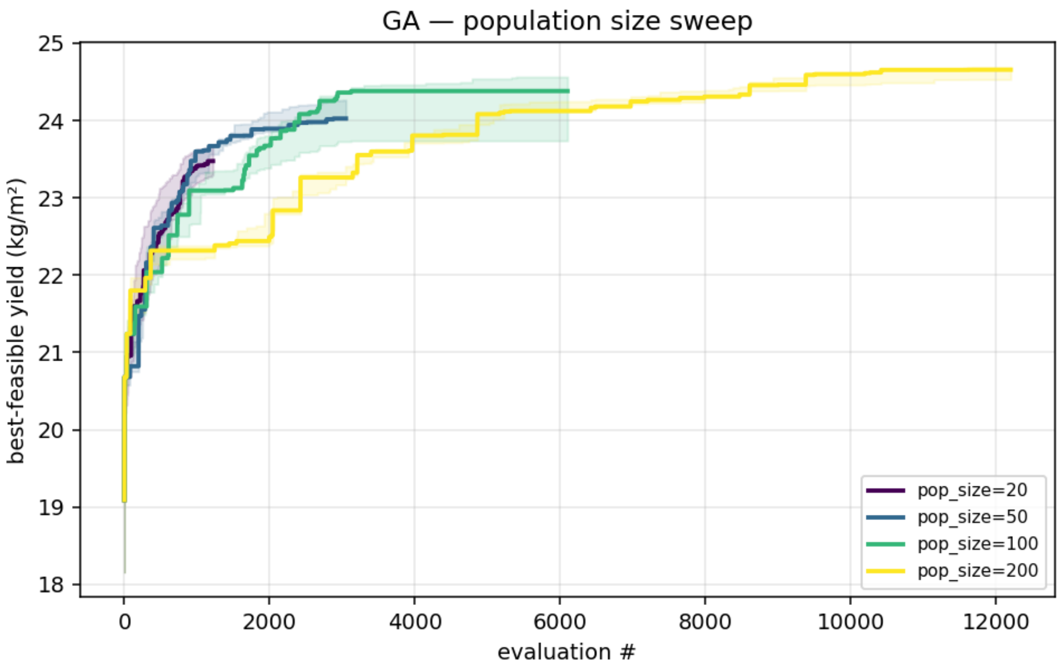

Figure 5: Genetic Algorithm population-size sweep on tomato. Larger populations improve final best-feasible yield under the current cheap-surrogate setting, although they also require more total evaluations before converging. This supports the interpretation that broader exploration is helpful in the 14-dimensional constrained search space.

B.3 Structure of the Optimized Solutions

Figure 6 compares the best decision vectors found for each crop and algorithm after normalizing each variable to its box bounds.

Figure 7 shows a local sensitivity analysis around the best tomato solution, and Figure 8 shows the operating-cost composition of the best feasible solutions.

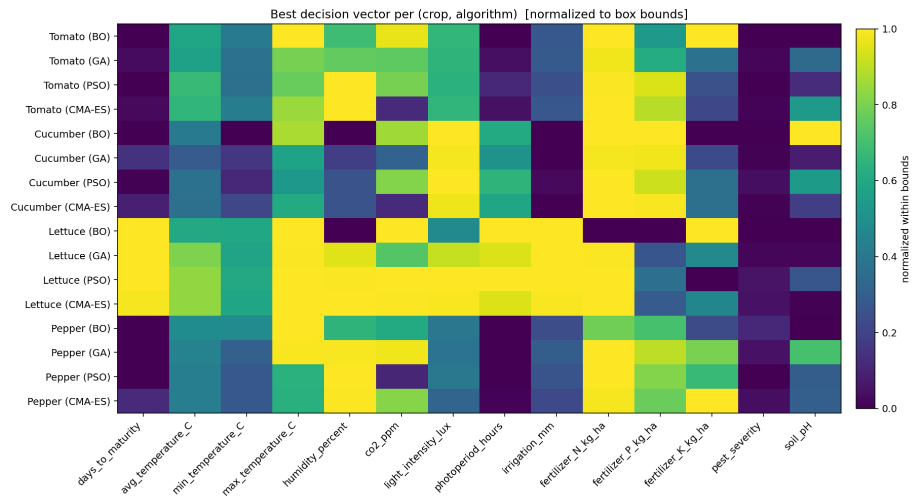

Figure 6: Best decision vector per crop and algorithm, normalized to each variable's box bounds. Several consistent patterns appear across methods, including high light intensity, high nitrogen, and low pest severity, while other variables such as fertilizer K and photoperiod vary more strongly across algorithms.

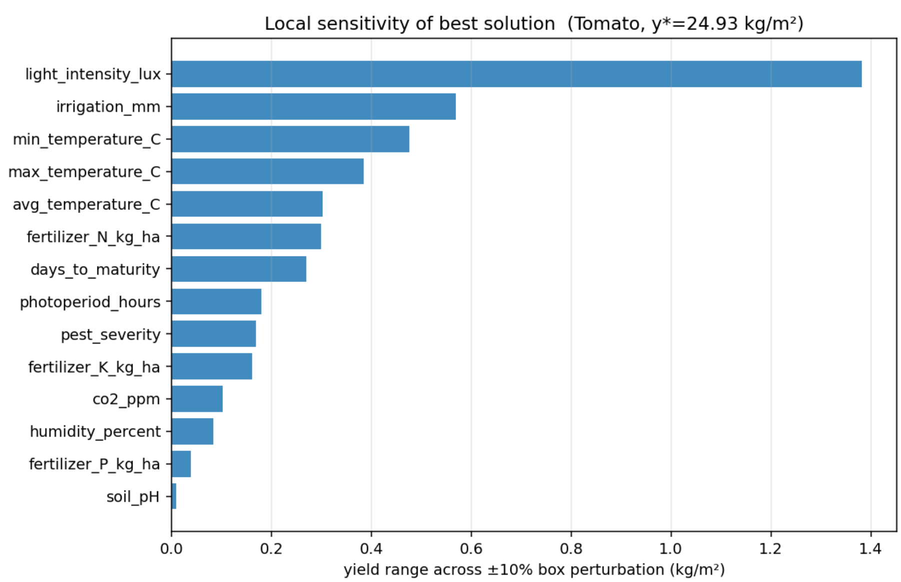

Figure 7: Local one-at-a-time sensitivity analysis around the best tomato solution. Light intensity is the most sensitive variable, followed by irrigation and temperature-related variables, while soil pH and phosphorus have relatively small local effects near the optimum.

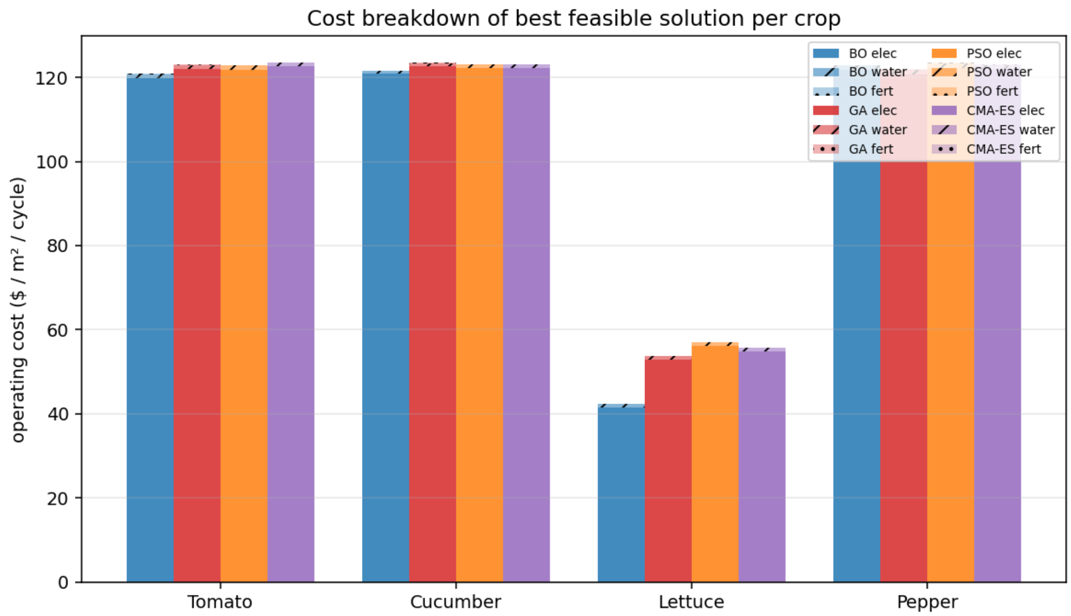

Figure 8: Operating-cost breakdown of the best feasible solution for each crop and algorithm. Electricity dominates total operating cost across all crops, while water and fertilizer contribute much smaller shares. This helps explain why many high-yield solutions concentrate near the electricity budget boundary.

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